Ring-LWE and Ideal Lattice Cryptography

Ring-LWE 與理想格密碼學

Algebraic Number Theory Meets Post-Quantum Security | 代數數論與後量子安全的交匯

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研究動機 (Motivation)

Classical public-key cryptography is provably broken by quantum computation. The two pillars of contemporary public-key infrastructure are RSA (security based on the integer factoring problem) and ECC (security based on the elliptic-curve discrete logarithm problem). **Shor's algorithm** (P. Shor, FOCS 1994) solves both in *quantum polynomial time* $\widetilde{O}((\log N)^3)$, whereas the best known classical algorithms (GNFS, Pollard rho) require sub-exponential time $\exp\big(\widetilde{O}((\log N)^{1/3})\big)$. Once a sufficiently large fault-tolerant quantum computer exists, RSA, Diffie–Hellman, and ECC all collapse simultaneously.

傳統公鑰系統 RSA 與 ECC 之安全性分別建基於**整數分解**與**橢圓曲線離散對數**。**Shor 演算法**（Shor, 1994）將兩者於量子電腦上同時化為多項式時間 $\widetilde{O}((\log N)^3)$，較古典最佳演算法 GNFS 之亞指數複雜度顯著加速。一旦容錯量子計算機問世，整個公鑰基礎設施將同時失守。

Lattice-based cryptography provides the leading post-quantum alternative: its security reduces, in the worst case, to geometric problems on lattices (*shortest vector*, *closest vector*) for which no quantum polynomial-time algorithm is known. **Ring-LWE** (Lyubashevsky–Peikert–Regev 2010) further exploits the algebraic structure of number rings to reduce the public key from $O(n^2)$ to $O(n)$ and the multiplication cost from $O(n^2)$ to $O(n \log n)$ via the number-theoretic transform.

格密碼學的安全性可歸約至格上最差情形的 SVP/CVP，迄今無已知量子多項式時間演算法。Ring-LWE 進一步以數環的代數結構，將公鑰自 $O(n^2)$ 壓縮至 $O(n)$、乘法自 $O(n^2)$ 加速至 $O(n \log n)$，兼得可證明安全與實用效率。

報告架構（四步驟）(Four-Step Structure)

Step 1 — Lattices and Geometry of Numbers（格與幾何數論）

A **lattice** of rank n is a discrete subgroup $\mathcal{L} = \bigoplus_{i=1}^{n} \mathbb{Z}\, b_i \subset \mathbb{R}^n$ generated by linearly independent $b_1, \ldots, b_n \in \mathbb{R}^n$. The basis matrix $B = [b_1 \mid \cdots \mid b_n]$ defines the *covolume* $\det(\mathcal{L}) = |\det B|$, an invariant of $\mathcal{L}$. The successive minima are

$$\lambda_i(\mathcal{L}) := \inf\{r > 0 : \dim \operatorname{span}\big(\mathcal{L} \cap \overline{B(0,r)}\big) \geq i\}, \quad i = 1, \ldots, n.$$

Three central hard problems: SVP_γ (find $v \in \mathcal{L} \setminus \{0\}$ with $\|v\| \leq \gamma\, \lambda_1$), GapSVP_γ (decide $\lambda_1 \leq 1$ vs. $> \gamma$), and BDD_d (decode $t \in \mathbb{R}^n$ within $d \cdot \lambda_1/2$). The best classical and quantum algorithms at small γ require time $2^{\Theta(n)}$.

Minkowski's first theorem. For any centrally symmetric convex body $S \subset \mathbb{R}^n$ with $\mathrm{vol}(S) > 2^n \det(\mathcal{L})$, S contains a nonzero lattice point. Applying this to a centered cube yields

$$\lambda_1(\mathcal{L}) \leq \sqrt{n} \cdot \det(\mathcal{L})^{1/n},$$

sharpened via the Hermite constant $\gamma_n \leq \frac{2n}{\pi e} + o(n)$.

格為 $\mathbb{R}^n$ 中的離散加法子群，其覆積 $\det(\mathcal{L})$ 與相繼極小 λ_i 為核心不變量。SVP_γ、GapSVP_γ、BDD_d 為主要困難問題，目前最佳演算法皆為指數時間。Minkowski 凸體定理直接給出上式，建立**幾何（covolume）與算術（最短向量）的橋樑**——即 **Course Objective 5** 之核心。

Step 2 — Number Rings as Ideal Lattices（數環作為理想格）

Setup. Fix $m = 2^k$ with $k \geq 2$ and let $K = \mathbb{Q}(\zeta_m)$ be the m-th cyclotomic field. The field degree is

$$n := [K : \mathbb{Q}] = \varphi(m) = 2^{k-1}, \qquad \Phi_m(x) = x^n + 1.$$

The ring of integers $\mathcal{O}_K = \mathbb{Z}[\zeta_m]$ is a Dedekind domain (**Obj. 1, 2**); every nonzero ideal admits a unique factorisation $\mathfrak{a} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r}$.

Splitting of primes (Obj. 3). For a rational prime $p \nmid m$, let f be the order of p modulo m. Then $p\mathcal{O}_K = \mathfrak{p}_1 \cdots \mathfrak{p}_g$ splits into $g = n/f$ distinct primes of inertia degree f. For $p = 2$ (the unique ramified prime), $2\mathcal{O}_K = (1 - \zeta_m)^n$ is *totally ramified*.

Trace, norm, discriminant (Obj. 4). For $\alpha \in K$, $\mathrm{Tr}_{K/\mathbb{Q}}(\alpha) = \sum_i \sigma_i(\alpha)$ and $\mathrm{N}_{K/\mathbb{Q}}(\alpha) = \prod_i \sigma_i(\alpha)$. The *different* $\mathfrak{d}_{K/\mathbb{Q}}^{-1} = \{\alpha \in K : \mathrm{Tr}(\alpha \mathcal{O}_K) \subseteq \mathbb{Z}\}$ satisfies $\Delta_K = \mathrm{N}_{K/\mathbb{Q}}(\mathfrak{d}_{K/\mathbb{Q}})$. For

$m = 2^k$, $|\Delta_K| = 2^{n(k-1)}$.

Canonical (Minkowski) embedding. Since $r_1 = 0, r_2 = n/2$,

$$\sigma\colon K \hookrightarrow K_{\mathbb{R}} := K \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{C}^{n/2} \cong \mathbb{R}^n, \qquad \alpha \mapsto \big(\sqrt{2}\,\Re\sigma_1(\alpha), \sqrt{2}\,\Im\sigma_1(\alpha), \ldots\big).$$

Under this Minkowski normalisation, every nonzero ideal $\mathfrak{a} \subseteq \mathcal{O}_K$ becomes a full-rank **ideal lattice** with covolume

$$\det\big(\sigma(\mathfrak{a})\big) = \mathrm{N}(\mathfrak{a})\,\sqrt{|\Delta_K|}.$$

Worked example ($m = 8$, $n = 4$). $K = \mathbb{Q}(\zeta_8)$, $\Phi_8(x) = x^4 + 1$, $|\Delta_K| = 2^{4\cdot 2} = 256$, $\sqrt{|\Delta_K|} = 16$. For $q = 17 \equiv 1 \pmod 8$: $f = 1$, so $17\mathcal{O}_K = \mathfrak{q}_1\mathfrak{q}_2\mathfrak{q}_3\mathfrak{q}_4$ splits completely — foreshadowing the NTT factorisation of R_{17} in Step 3.

數環 $\mathcal{O}_K$ 為 Dedekind domain，理想可唯一分解為質理想之積，貫穿 **Obj. 1–4**。Minkowski 嵌入下理想格之覆積為 $\mathrm{N}(\mathfrak{a})\sqrt{|\Delta_K|}$，此即**把代數結構翻譯為幾何結構**的關鍵公式。$m = 8$ 時 $|\Delta_K| = 256$、$q = 17$ 完全分裂為四個質理想，預示 Step 3 的 NTT。

Step 3 — LWE and Ring-LWE（LWE 與其環版本）

LWE (Regev, STOC 2005 / JACM 2009): fix n, q and an error distribution χ on $\mathbb{Z}_q$ (typically discrete Gaussian of width αq with $\alpha < 1/\sqrt{n}$). For fixed $s \in \mathbb{Z}_q^n$, an LWE sample is

$$(a,\ \langle a, s\rangle + e \bmod q) \in \mathbb{Z}_q^n \times \mathbb{Z}_q, \qquad a \leftarrow U(\mathbb{Z}_q^n),\ e \leftarrow \chi.$$

Regev's quantum reduction: $\mathrm{LWE}_{n,q,\chi} \longleftarrow \mathrm{GapSVP}_\gamma,\ \mathrm{SIVP}_\gamma$ on arbitrary lattices, $\gamma = \widetilde{O}(n/\alpha)$. Drawback: the public key $A \in \mathbb{Z}_q^{n\times m}$ has size $O(n^2)$, with $O(n^2)$ multiplication.

Ring-LWE (Lyubashevsky–Peikert–Regev, EUROCRYPT 2010 / JACM 2013): replace $\mathbb{Z}^n$ by the cyclotomic ring of integers

$$R := \mathcal{O}_K = \mathbb{Z}[x]/\Phi_m(x) = \mathbb{Z}[x]/(x^n + 1), \qquad R_q := R/qR.$$

Let $R^\vee := \mathfrak{d}_{K/\mathbb{Q}}^{-1}$ be the codifferent (**Obj. 4**). For secret $s \in R_q^\vee$ and Gaussian error e on $K_{\mathbb{R}}$, a Ring-LWE sample is $(a,\ a{\cdot}s{+}e \bmod qR^\vee) \in R_q \times R_q^\vee$. **Reduction:** Ring-LWE $\longleftarrow$ approx-SVP on *ideal lattices* in R (quantum, polynomial-time). For power-of-two cyclotomics, $R^\vee = \frac{1}{n}R$ is a scalar dilation, so the dual collapses to a rescaling —this is the *simplified* Ring-LWE used in implementations.

The role of prime splitting (Obj. 3). Choose q prime with $q \equiv 1 \pmod m$. Then $\Phi_m(x)$ has

n distinct roots in $\mathbb{Z}_q$:

$$\Phi_m(x) \equiv \prod_{i=1}^{n}(x - \omega_i) \pmod q, \qquad R_q \cong \prod_{i=1}^{n} \mathbb{Z}_q \ \text{(CRT)}.$$

This isomorphism is precisely the **Number-Theoretic Transform (NTT)**, computable in $O(n \log n)$. Multiplication in R_q becomes pointwise after NTT, mirroring convolution-via-FFT. Geometrically, $q\mathcal{O}_K$ splits completely into n distinct primes.

Negacyclic convolution. Since $\Phi_m(x) = x^n + 1$, $x^n \equiv -1$ in R_q, so $(a * b)_k = \sum_{i+j=k} a_i b_j - \sum_{i+j=k+n} a_i b_j$, $0 \le k < n$. The Ring-LWE public key $(a, b) \in R_q \times R_q^\vee$ has total size $O(n)$ — a factor-n saving.

LWE 樣本 $(a, \langle a, s\rangle + e)$ 之安全性歸約至任意格上的 worst-case SVP/SIVP。Ring-LWE 將 $\mathbb{Z}^n$ 換為分圓整數環 $R = \mathcal{O}_K$，安全性歸約至**理想格**上的 approx-SVP；逆差理想 $R^\vee = \mathfrak{d}_{K/\mathbb{Q}}^{-1}$ 對應 **Obj. 4**，質理想完全分裂（**Obj. 3**）對應 NTT 之 CRT 同構。公鑰自 $O(n^2)$ 縮為 $O(n)$，乘法自 $O(n^2)$ 加速至 $O(n \log n)$。

Step 4 — Applications（應用）

日常加密連線（HTTPS、行動銀行、訊息、VPN）皆仰賴 RSA/ECC，量子電腦出現將同時失效；且 **今日攔截、未來解密**的威脅已存在。NIST 於 2024 年 8 月公布首批後量子標準，其中 ML-KEM（Kyber）取代 RSA/DH 金鑰交換、ML-DSA（Dilithium）取代 RSA/ECDSA 簽章，皆建立於 Ring-LWE 之模版本（Module-LWE）。

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